

Besides efficient insurance claims processing, payers can use predictive analytics to determine at-risk claims. They can also find the best providers for specific health conditions, enabling patients to get better returns on their claims.

Technologies to achieve the feat

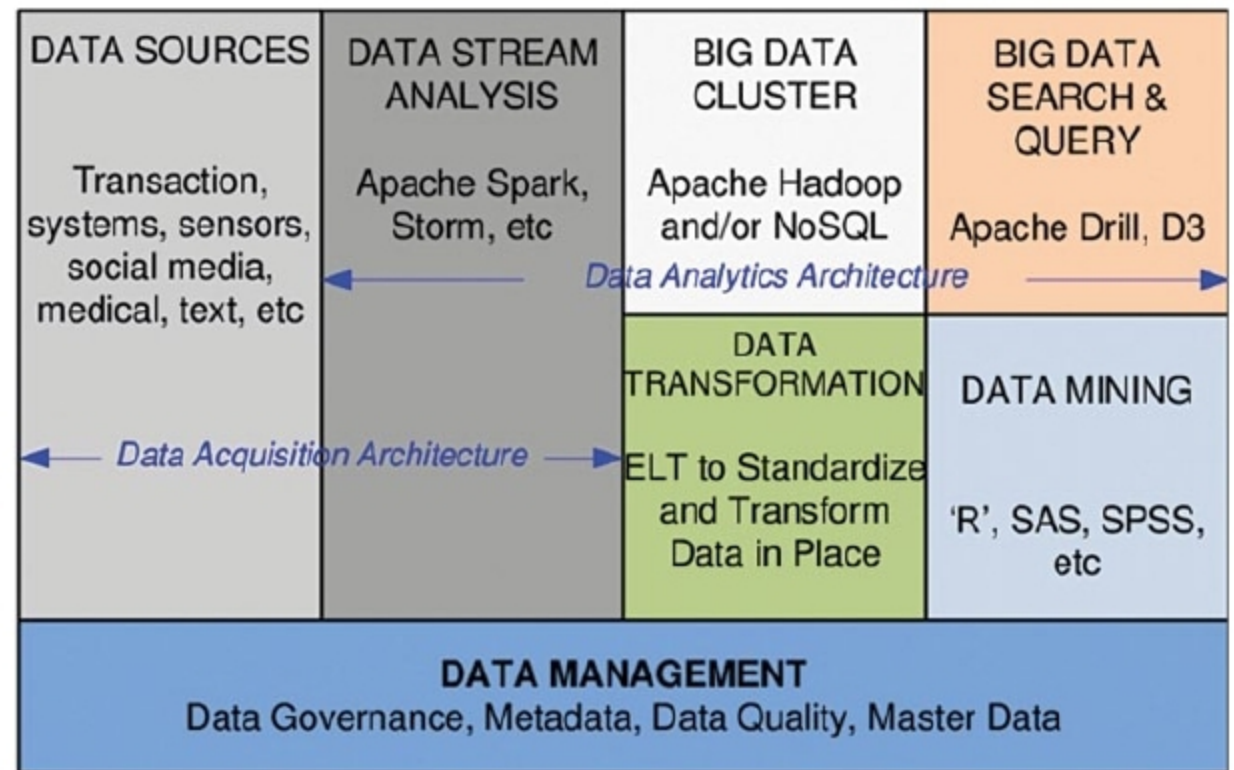
Formulation of a well developed data analytics solution is crucial for preventive, predictive, precise and personalised healthcare. Cloud services can economically and easily manage huge volumes of data and share information across different systems, making data analytics possible.

For scalable implementation, it is always a good idea to process all the information to extract the important data while not missing out on any detail, which can be then integrated into the system to gain meaningful insights. This is where the idea of decentralised processing, that is, edge cloud computing comes into play. Unlike traditional cloud, the information is stored locally, increasing network performance and reducing response times.

AI and ML can be used to analyse both structured (such as ontologies) and unstructured data to support medical professionals in decision making, and related policy decisions by providing a holistic view of a particular situation. The Pittsburgh Health Data Alliance started working with Amazon Web Services and utilising its ML research program to boost innovation and revolutionise disease treatment with data last year.

Apache spark is a popular framework with built-in fault tolerance for iterative and interactive processes in Big Data analysis. It is written in Scala and can be integrated with Hadoop. PySpark API allows using spark framework with Python, a popular language among developers for ML due to its various advantages like flexibility, robustness and ease of implementation.

For predictive modeling, the inputs and outputs for the model can be discrete or continuous, and are decided based on the type of problem. Neural networks are trained with past data available to under-



Big data analytics architecture framework (Credit: www.researchgate.net)

stand the patterns and trends in the gathered datasets, which is known as data mining. Based on similarities in characteristics with the data of former patients, new ones can be more effectively treated.

It is important to assess the performance of a model to understand the improvements that can be made.

Impact and challenges

One of the best ways to understand the impact of health data analytics is from the response of Taiwan to coronavirus pandemic (COVID-19). According to a Stanford health report, the state integrated its national health insurance database with its immigration and customs database to create Big Data for analytics. QR code scanning and online reporting were also allowed.

This allowed identification of susceptible cases who had recent travel history to high-risk areas and were visiting clinics with symptoms through real-time tracking. People who had travelled to high-risk areas were tracked through their phones to make sure they stayed at home during the quarantine period. This helped them nip the disease in the bud.

Healthcare data involves numerous complexities that need to be tackled. For instance, medical data comes from different sources and has to comply with regulations set by different state governments and other administrative departments. There has to be a common standard when it

comes to sharing patient information.

In 2019, a former patient at University of Chicago Medical Center filed a lawsuit against the institution's partnership with Google to improve predictive analysis, stating that it violated Health Insurance Portability and Accountability Act (HIPAA) passed in 1996 in the US by recording the entry and exit dates of patients. The lawsuit was dismissed later on.

Technically too, developing infrastructure to interface datasets from different data providers on such a large scale is not easy. Security loopholes in the system can have drastic implications. So, advanced security measures like firewalls and encryption algorithms are essential to avoid risks and maintain the trust of patients.

Due to such issues, cloud alternatives are considered a safe option to reduce vulnerability by many companies. Corporate Health International (CHI) is one such organisation that is utilising the computational efficiency provided by hardware and software stack from Intel. Servers running Intel Xeon are used for both data processing and AI development.

In spite of realising its potential, healthcare organisations in many countries are still not actively using analytics due to these issues. Developments are ongoing to resolve such problems and avail the untapped benefits from analytics that are widespread in other industries already. **EFY**